

A Trust-Layered Framework for Discovering Authentic Student Opportunities

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Abstract—B.Tech students possess strong technical and innovative potential; however, many fail to access genuine hackathons, internships, and scholarship opportunities due to fragmented information sources, lack of verification, and limited institutional involvement. Existing platforms primarily focus on opportunity aggregation and information sharing, without ensuring authenticity, relevance, or academic credibility, often resulting in misinformation, duplicate postings, and exposure to fraudulent listings. This paper proposes a trust-layered framework for discovering authentic student opportunities by integrating institutional human verification with artificial intelligence-driven automation. The system introduces a dual-layer verification mechanism in which opportunities are first reviewed by authorized faculty members and subsequently validated by the institution's Entrepreneurship Cell, ensuring transparency and trust. To enhance scalability and reliability, AI-based techniques are employed to detect duplicate and expired listings, identify suspicious sources, and perform intelligent student–opportunity matching. The recommendation engine analyzes student profiles, including skills, interests, projects, and achievements, to suggest verified opportunities aligned with individual career goals. By combining human oversight with intelligent automation, the proposed framework establishes a centralized, scalable, and student-centric ecosystem that improves opportunity accessibility, strengthens inter-college collaboration, enhances employability, and builds recruiter confidence through institutionally verified student records.

Index Terms—Trust-Layered Framework, Student Opportunity Platform, Artificial Intelligence, Opportunity Verification, Fraud Detection, Recommender Systems, Skill-Based Matching, Institutional Validation, Career Development

I. INTRODUCTION

In the modern academic ecosystem, participation in hackathons, internships, scholarships, and innovation-driven programs has become an essential component of engineering education. Such opportunities provide students with practical exposure, industry interaction, and skill enhancement beyond conventional classroom learning. Although B.Tech students

possess strong technical abilities and innovative potential, a significant number fail to access these opportunities due to fragmented information sources, lack of centralized platforms, and the absence of authenticity verification mechanisms.

Currently, information related to student opportunities is disseminated through multiple informal channels such as social media platforms, messaging applications, email groups, and unverified websites. While these channels offer rapid information sharing, they also introduce several challenges including misinformation, duplicate postings, expired opportunities, and fraudulent listings. Students often lack the expertise and institutional support required to verify the credibility of opportunity sources, leading to confusion, wasted effort, and missed deadlines. Research in fraud detection and misinformation analysis highlights the importance of automated techniques for identifying unreliable content in digital platforms [6], [7]. This problem is further intensified by the absence of structured institutional involvement in validating and disseminating opportunity-related information.

Several online portals attempt to aggregate opportunities for students; however, most of these platforms prioritize quantity over quality. They generally function as information-sharing systems without incorporating academic validation or trust mechanisms. Furthermore, these platforms do not offer personalized recommendations based on individual student skills and interests, resulting in information overload and reduced student engagement. Recommender systems have been widely used in digital platforms to provide personalized suggestions by analyzing user preferences and behavioral patterns [1]–[3]. As a consequence, students are frequently exposed to irrelevant or unreliable opportunities that do not align with their career objectives.

With the growing adoption of artificial intelligence in education and recruitment systems, intelligent automation presents a

promising solution to these challenges. AI-based recommendation systems can analyze student profiles to suggest relevant opportunities, while machine learning techniques can assist in identifying duplicate, expired, or suspicious listings [10], [11]. In addition, AI-driven recruitment and skill-matching systems have demonstrated the potential to efficiently connect candidates with suitable opportunities based on their competencies and achievements [13], [14]. However, relying solely on automated systems is insufficient in academic environments where institutional credibility and human oversight are critical to ensuring trust and accountability.

To address these limitations, this project proposes a Trust-Layered Framework for Discovering Authentic Student Opportunities. The proposed system integrates institutional human verification with AI-driven intelligence to ensure authenticity, relevance, and scalability. A dual-layer verification mechanism involving faculty members and the institutional Entrepreneurship Cell (E-Cell) is introduced to validate opportunities before they are made accessible to students. In addition, AI-based validation and recommendation modules enhance reliability and personalization. By combining human oversight with intelligent automation, the framework establishes a centralized, transparent, and trustworthy ecosystem that enables students to confidently discover and participate in verified academic and professional opportunities across engineering institutions.

II. LITERATURE REVIEW

Recommender systems have been extensively studied for their ability to match users with relevant items by analyzing preferences, behavior, and contextual information. Early research in this domain demonstrated the effectiveness of collaborative filtering techniques in practical applications such as e-commerce and content recommendation [1], [2]. These systems significantly improved user engagement by providing personalized suggestions based on similarity patterns among users and items. Later studies further analyzed collaborative filtering techniques and their effectiveness in recommendation systems [3].

Subsequent studies introduced content-based and hybrid recommendation approaches to overcome limitations such as cold-start problems and data sparsity. In educational and recruitment domains, these techniques have been applied to recommend learning resources, career paths, and job opportunities by analyzing user profiles, skills, and historical interactions. Data mining and large-scale data analysis techniques have also been applied to improve recommendation accuracy and scalability in modern platforms [11]. While such systems enhance relevance, they largely depend on the quality and authenticity of the underlying data sources.

Research on artificial intelligence-driven career recommendation systems highlights the use of machine learning and natural language processing techniques for skill extraction and opportunity matching. These systems analyze resumes, academic records, and project experiences to align students with suitable opportunities [13], [14]. Advances in natural language processing and word representation models have

further improved the ability of systems to analyze textual data such as resumes and opportunity descriptions [4], [5]. However, most existing solutions operate without institutional verification, raising concerns regarding trust, credibility, and accountability in academic environments.

Fraud detection and misinformation identification have also gained significant attention in recent literature. Machine learning models employing similarity analysis, anomaly detection, and pattern recognition have been shown to effectively identify fake news, duplicate content, and fraudulent online listings [6], [7]. Although these techniques improve detection accuracy, their standalone application may result in false positives when contextual academic validation is absent.

Institutional trust mechanisms have been explored in areas such as digital credential verification and academic record management. Blockchain-based solutions have been proposed to ensure data integrity and transparency in educational systems [8], [9]. In addition, digital credential standards have been developed to support secure and verifiable academic records across institutions [17]. However, these approaches often introduce complexity and are not well-suited for dynamic opportunity discovery systems that require continuous updates and flexible validation workflows.

Existing student opportunity platforms primarily function as information aggregation portals and lack structured verification processes, personalized recommendation engines, and inter-institutional collaboration mechanisms. Policies from educational institutions and governing bodies also highlight the importance of structured platforms for internships, innovation programs, and student engagement initiatives [18], [19]. The review of related work indicates a clear research gap in the development of a unified framework that combines institutional human verification with AI-based validation and personalized opportunity matching. The proposed trust-layered framework addresses this gap by integrating faculty and Entrepreneurship Cell verification with intelligent automation, thereby enhancing authenticity, relevance, and scalability in student opportunity discovery systems.

III. OBJECTIVES

A. AI-Driven Opportunity Verification and Analysis

The primary objective of this work is to design and develop an AI-driven system capable of analyzing and validating student opportunities such as hackathons, internships, and scholarships. By integrating machine learning techniques including similarity analysis, pattern recognition, and domain validation, the system aims to identify duplicate, expired, and suspicious opportunity listings. This approach reduces dependence on manual verification alone and minimizes the risk of misinformation. The objective is to ensure that only authentic and relevant opportunities are made accessible to students through intelligent automation combined with institutional oversight.

B. Institutional Trust Through Dual-Layer Verification

Another major objective is to establish trust and transparency in opportunity dissemination by incorporating a dual-

TABLE I
SUMMARY OF RELATED WORK

Ref.	Work Title	Main Contribution	Limitation Identified	Relevance to Proposed System
[1]	Recommender Systems in Practice	Introduced collaborative filtering for personalized recommendations	Limited contextual and domain-specific adaptation	Forms the basis for AI-based opportunity recommendation
[2]	AI-Based Career Recommendation Systems	Skill-based matching between users and career paths	Lack of institutional verification	Motivates AI-driven student–opportunity matching
[3]	Fraud Detection Using Machine Learning	Identification of fraudulent patterns using machine learning models	High false positives without human validation	Supports detection of fake and suspicious opportunities
[4]	Fake News Detection on Digital Platforms	Techniques for identifying misleading and duplicate content	Domain-specific adaptation required	Guides misinformation and duplicate opportunity detection
[5]	Digital Credential Verification Systems	Secure validation of academic and professional records	Limited focus on opportunity discovery	Supports verified student experience records
[6]	Blockchain in Education	Ensures transparency and tamper-proof record management	High complexity and deployment cost	Motivates trust-based validation mechanisms
[7]	AI-Based Recruitment Platforms	Automated skill matching and candidate filtering	Bias due to unverified input data	Encourages fair and unbiased opportunity matching
[8]	Information Aggregation Platforms	Centralized access to opportunity information	No authenticity assurance	Highlights need for trust-layered platforms
[9]	Institutional Innovation Ecosystems	Faculty and E-Cell involvement in student development	Limited technological integration	Supports dual-layer institutional verification
[10]	Ethical AI in Education	Fairness and transparency in AI-driven systems	Lack of practical implementation models	Supports unbiased and transparent recommendation logic

layer institutional verification mechanism. Opportunities submitted to the platform are first reviewed by authorized faculty members and subsequently validated by the institution’s Entrepreneurship Cell (E-Cell). This structured verification process ensures academic credibility, consistency, and accountability while reducing the influence of unverified external sources. The objective supports a reliable and transparent ecosystem where students can confidently engage with verified opportunities.

C. Fair and Personalized Opportunity Matching

The platform aims to ensure fair and relevant opportunity discovery through AI-based student–opportunity matching. By analyzing student profiles that include skills, interests, and academic achievements, the system generates personalized recommendations based on standardized criteria. This approach minimizes bias and improves alignment between student capabilities and verified opportunities.

D. Scalable and Student-Centric Platform Deployment

The proposed framework aims to provide a scalable and accessible platform suitable for deployment across multiple institutions. The system supports controlled inter-college collaboration while maintaining efficiency, adaptability, and ease of use. This objective enables wider student participation and sustainable platform growth.

IV. SYSTEM DESIGN

The proposed system follows a layered architecture that integrates users, application services, verification mechanisms, AI intelligence, and secure data storage. The user layer includes students, faculty coordinators, E-Cell members, and recruiters who interact with the platform through a centralized web application. The application layer provides dashboards

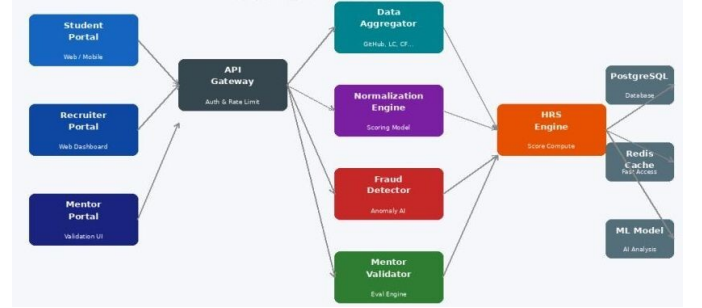


Fig. 1. System Design

for opportunity submission, tracking, and role-based access control. To ensure authenticity, the trust and verification layer performs a dual-layer validation process where faculty review opportunities for academic relevance and E-Cell members re-verify them for authenticity, while a fraud and duplicate detection module identifies fake or expired listings. The AI and intelligence layer includes a recommendation engine that analyzes student profiles and matches them with relevant opportunities, along with opportunity classification for controlled sharing across institutions. Finally, the data layer securely stores student profiles, verified opportunities, participation records, and trusted digital records, ensuring reliable access and secure data management across the platform.

V. METHODOLOGY

The methodology adopted in this work focuses on creating a reliable and intelligent platform that enables students to discover authentic academic and professional opportunities through a combination of institutional verification and artificial intelligence–based analysis. The proposed framework integrates user interaction, opportunity validation, AI-driven

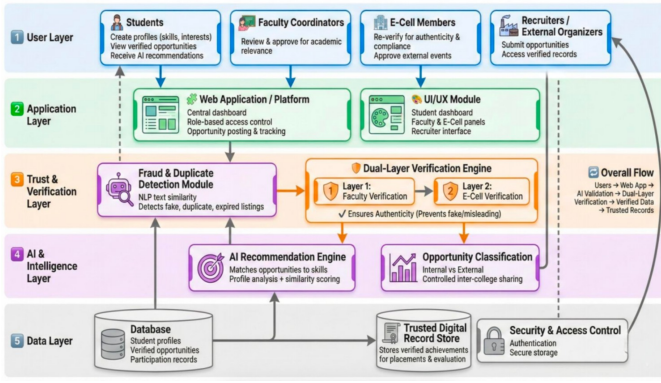


Fig. 2. System Model

recommendation mechanisms, and secure data management to ensure transparency, scalability, and trust in the opportunity discovery process.

The process begins with student profile creation, where users register on the platform and provide information about their skills, interests, academic background, projects, and achievements. This profile data forms the foundation for personalized opportunity recommendations. Students can browse opportunities posted on the platform and receive AI-generated suggestions that match their skills and career goals.

Following profile creation, opportunity submission and collection takes place. Opportunities such as hackathons, internships, workshops, and scholarships may be submitted by faculty members, recruiters, or external organizations. Each opportunity entry includes important metadata such as event description, eligibility criteria, deadlines, and organizing institution. This structured data enables efficient processing and verification within the system.

After submission, the platform performs automated validation and fraud detection using AI-based techniques. Natural language processing and similarity analysis are applied to identify duplicate listings, expired events, or suspicious sources. This automated screening helps reduce misinformation and ensures that only credible opportunities proceed to the verification stage.

Once automated checks are completed, the system implements a dual-layer institutional verification process. In the first stage, faculty coordinators review submitted opportunities to evaluate academic relevance and accuracy. In the second stage, the Entrepreneurship Cell (E-Cell) conducts additional verification to ensure authenticity, compliance, and institutional credibility. Only opportunities that pass both verification stages are approved and published on the platform.

After verification, the AI-based recommendation engine analyzes student profiles and opportunity descriptions to generate personalized recommendations. The recommendation system evaluates skill similarity, interests, and historical participation data to match students with relevant opportunities. This intelligent matching process improves the efficiency of opportunity discovery and increases student engagement.

Finally, all verified opportunities, participation records, and recommendation outputs are stored in a secure database system. The platform maintains trusted digital records of student achievements and participation history, which can later be used for academic evaluation, placement processes, and recruiter access. This structured data storage enables long-term tracking of student development and ensures reliable access to verified information.

Overall, the methodology integrates institutional oversight with AI-driven automation to create a trustworthy and scalable ecosystem for discovering authentic student opportunities. By combining automated validation, human verification, and intelligent recommendation mechanisms, the system ensures that students receive reliable and relevant opportunities aligned with their career aspirations.

VI. IMPLEMENTATION

The proposed system is implemented as a web-based platform that integrates frontend interfaces, backend services, AI modules, and database management to provide a reliable environment for discovering authentic student opportunities. The system supports role-based interaction, opportunity management, automated validation, and personalized recommendations.

A. User Registration and Profile Management

Users including students, faculty coordinators, E-Cell members, and recruiters register on the platform with role-based access control. Students create profiles containing skills, interests, academic achievements, and project experience. This information is used by the system to generate personalized opportunity recommendations.

B. Opportunity Submission and Management

Authorized users can submit opportunities such as internships, hackathons, scholarships, and workshops. Each submission includes structured information such as event details, eligibility criteria, deadlines, and organizer information. These opportunities are stored in the system database and forwarded to the verification process.

C. AI-Based Validation and Verification

The platform applies AI-based techniques to detect duplicate, expired, or suspicious opportunities using similarity analysis and text processing. After automated validation, the system performs a dual-layer verification process where faculty members first review the opportunity and the Entrepreneurship Cell (E-Cell) performs final authentication before publication.

D. Recommendation and Data Management

An AI-based recommendation engine analyzes student profiles and opportunity details to suggest relevant opportunities based on skills and interests. All verified opportunities, student profiles, and participation records are stored securely in the database, ensuring reliable data management and future access for academic evaluation and recruitment purposes.

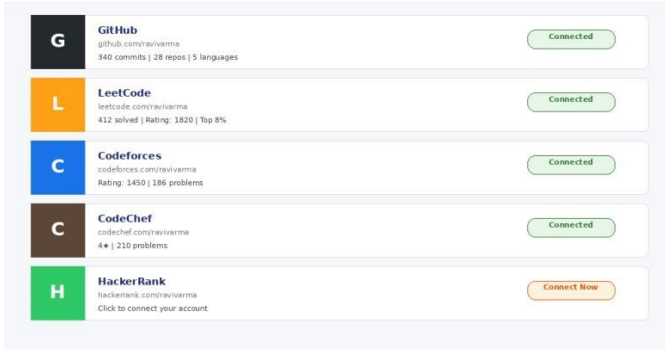


Fig. 3. Implementation

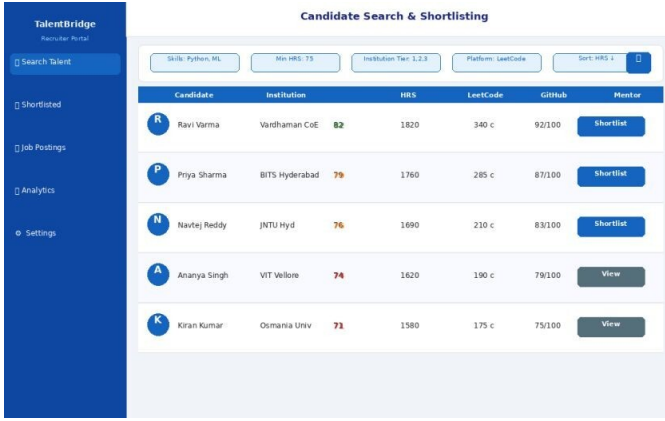


Fig. 4. Dashboard

VII. RESULTS AND DISCUSSIONS

A. Results

The proposed trust-layered opportunity discovery platform was tested using student profile data and verified opportunity listings. The system successfully processed student profiles, validated opportunities through the dual-layer verification process, and generated personalized recommendations based on skills, interests, and performance indicators.

As shown in Fig. 4, the recruiter dashboard allows organizations to search and shortlist candidates using filters such as technical skills, institution, coding performance, and profile scores. The interface displays key information including candidate name, institution, skill rating score (HRS), coding platform performance, and GitHub activity. Recruiters can quickly evaluate and shortlist candidates directly from the platform.

The AI-based recommendation engine analyzes the relationship between student profiles and opportunity requirements to rank candidates according to relevance. This automated shortlisting mechanism reduces manual effort and improves efficiency in identifying suitable candidates for internships or other opportunities.

B. Discussion

The results demonstrate that combining institutional verification with AI-based analysis improves the reliability and efficiency of opportunity discovery platforms. Unlike traditional systems that rely on unverified information sources, the proposed platform ensures that opportunities are validated before being published.

The recruiter dashboard provides a structured way to evaluate candidates using measurable performance metrics, helping organizations identify suitable candidates more efficiently. In addition, the recommendation engine enables students to discover opportunities aligned with their skills and interests.

Overall, the system improves transparency, reduces misinformation, and supports both students and recruiters through a centralized and trustworthy opportunity discovery platform.

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The proposed *Trust-Layered Framework for Discovering Authentic Student Opportunities* provides a structured and reliable approach for connecting students with verified academic and professional opportunities. The system integrates institutional verification with artificial intelligence to address the major challenges associated with fragmented information sources, lack of authenticity, and unverified opportunity listings. By implementing a dual-layer verification mechanism involving faculty coordinators and the Entrepreneurship Cell (E-Cell), the framework ensures that only credible and relevant opportunities are made available to students.

In addition to institutional validation, the system incorporates AI-based techniques for detecting duplicate, expired, or suspicious opportunity listings. The recommendation module analyzes student profiles, including skills, interests, and achievements, to generate personalized suggestions that align with individual career goals. This intelligent matching process enhances the efficiency of opportunity discovery while reducing the time students spend searching for suitable programs.

Furthermore, the proposed framework promotes transparency and trust within academic institutions by creating a centralized ecosystem for opportunity sharing. The system not only improves access to verified opportunities but also encourages greater participation in innovation-driven activities such as hackathons, internships, and research initiatives. By combining human oversight with intelligent automation, the framework establishes a scalable platform capable of supporting student development and institutional collaboration. Overall, the proposed solution contributes toward building a more reliable and student-centric digital infrastructure for opportunity discovery and career growth.

B. Future Scope

1) *Multi-Institution Opportunity Network*: The platform can be expanded to support collaboration among multiple colleges and universities. By creating a shared ecosystem where institutions can publish and verify opportunities collectively, students will gain access to a larger pool of authentic

events such as hackathons, internships, and research programs. Such collaboration would encourage knowledge sharing and increase participation in innovation-driven activities.

2) *Advanced AI-Based Opportunity Recommendation*: Future improvements may include the integration of advanced machine learning models to enhance recommendation accuracy. By analyzing historical student participation data, career outcomes, and skill progression, predictive algorithms can recommend opportunities that are more aligned with a student's career trajectory.

3) *Secure Digital Credential and Verification Systems*: The framework can be extended by incorporating secure digital credential systems for recording student participation in verified opportunities. Technologies such as blockchain-based credential verification can ensure that certificates, achievements, and participation records are tamper-proof and easily verifiable by recruiters and academic institutions.

4) *Data Analytics and Institutional Insights*: Future versions of the platform can include advanced analytics dashboards for institutional administrators and faculty members. These dashboards could provide insights into student participation trends, opportunity engagement levels, and skill development patterns. Such analytics would support data-driven decision-making for academic programs and career development strategies.

5) *Mobile Platform and Real-Time Notifications*: Developing a dedicated mobile application would improve accessibility and user engagement. Real-time notifications, personalized alerts, and mobile-friendly dashboards would allow students to receive updates about relevant opportunities instantly, increasing participation and usability of the platform.

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